

To see what $f_x(0.3, 1)$ means, think about the function $f(x, 1)$. The graph of $f(x, 1)$ in Figure 14.19 is a snapshot of the string at the time $t = 1$. Thus, the derivative $f_x(0.3, 1)$ is the slope of the string at the point $x = 0.3$ at the instant when $t = 1$.

Similarly, taking the derivative of $f(x, t) = 0.003 \sin(\pi x) \sin(2765t)$ with respect to t , we get

$$f_t(x, t) = (0.003)(2765) \sin(\pi x) \cos(2765t) = 8.3 \sin(\pi x) \cos(2765t).$$

Since $f(x, t)$ is in meters and t is in seconds, the derivative $f_t(0.3, 1)$ is in m/sec. Thus, substituting $x = 0.3$ and $t = 1$,

$$f_t(0.3, 1) = 8.3 \sin(\pi(0.3)) \cos(2765(1)) \approx 6 \text{ m/sec}.$$

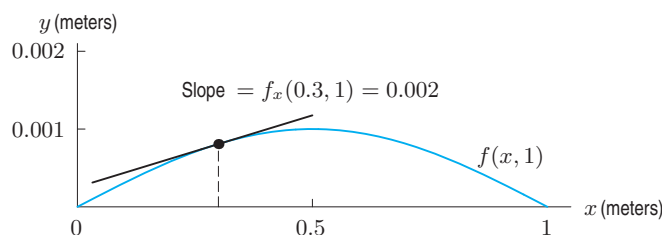


Figure 14.19: Graph of $f(x, 1)$: Snapshot of the shape of the string at $t = 1$ sec

To see what $f_t(0.3, 1)$ means, think about the function $f(0.3, t)$. The graph of $f(0.3, t)$ is a position versus time graph that tracks the up-and-down movement of the point on the string where $x = 0.3$. (See Figure 14.20.) The derivative $f_t(0.3, 1) = 6$ m/sec is the velocity of that point on the string at time $t = 1$. The fact that $f_t(0.3, 1)$ is positive indicates that the point is moving upward when $t = 1$.

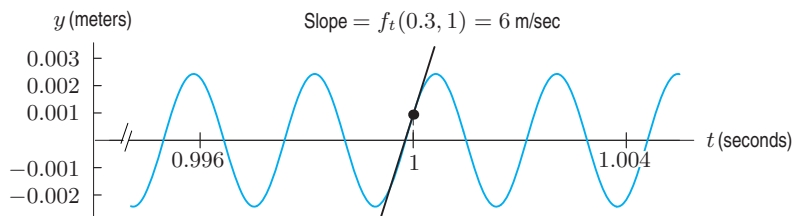


Figure 14.20: Graph of $f(0.3, t)$: Position versus time graph of the point $x = 0.3$ m from the end of the guitar string

Exercises and Problems for Section 14.2

Exercises

1. (a) If $f(x, y) = 2x^2 + xy + y^2$, approximate $f_y(3, 2)$ using $\Delta y = 0.01$.
(b) Find the exact value of $f_y(3, 2)$.

Find the partial derivatives in Exercises 2–40. Assume the variables are restricted to a domain on which the function is defined.

2. f_x and f_y if $f(x, y) = 5x^2y^3 + 8xy^2 - 3x^2$
3. $f_x(1, 2)$ and $f_y(1, 2)$ if $f(x, y) = x^3 + 3x^2y - 2y^2$

4. $\frac{\partial}{\partial y}(3x^5y^7 - 32x^4y^3 + 5xy)$

5. $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = (x^2 + x - y)^7$

6. f_x and f_y if $f(x, y) = A^\alpha x^{\alpha+\beta} y^{1-\alpha-\beta}$

7. f_x and f_y if $f(x, y) = \ln(x^{0.6}y^{0.4})$

8. z_x if $z = \frac{1}{2x^2ay} + \frac{3x^5abc}{y}$

9. z_x if $z = x^2y + 2x^5y$ 10. V_r if $V = \frac{1}{3}\pi r^2h$
11. $\frac{\partial}{\partial T} \left(\frac{2\pi r}{T} \right)$ 12. $\frac{\partial}{\partial x} (a\sqrt{x})$
13. $\frac{\partial}{\partial x} (xe^{\sqrt{xy}})$ 14. $\frac{\partial}{\partial t} e^{\sin(x+ct)}$
15. F_m if $F = mg$ 16. a_v if $a = \frac{v^2}{r}$
17. $\frac{\partial A}{\partial h}$ if $A = \frac{1}{2}(a+b)h$ 18. $\frac{\partial}{\partial m} \left(\frac{1}{2}mv^2 \right)$
19. $\frac{\partial}{\partial B} \left(\frac{1}{u_0} B^2 \right)$ 20. $\frac{\partial}{\partial r} \left(\frac{2\pi r}{v} \right)$
21. F_v if $F = \frac{mv^2}{r}$ 22. $\frac{\partial}{\partial v_0} (v_0 + at)$
23. $\frac{\partial F}{\partial m_2}$ if $F = \frac{Gm_1m_2}{r^2}$
24. $\frac{\partial}{\partial x} \left(\frac{1}{a} e^{-x^2/a^2} \right)$
25. $\frac{\partial}{\partial a} \left(\frac{1}{a} e^{-x^2/a^2} \right)$
26. f_x if $f(x, y) = e^{xy}(\ln y)$
27. $\frac{\partial}{\partial t} (v_0t + \frac{1}{2}at^2)$
28. $\frac{\partial}{\partial \theta} (\sin(\pi\theta\phi) + \ln(\theta^2 + \phi))$
29. $\frac{\partial}{\partial M} \left(\frac{2\pi r^{3/2}}{\sqrt{GM}} \right)$
30. f_a if $f(a, b) = e^a \sin(a+b)$
31. z_x if $z = \sin(5x^3y - 3xy^2)$
32. g_x if $g(x, y) = \ln(ye^{xy})$
33. F_L if $F(L, K) = 3\sqrt{LK}$
34. $\frac{\partial V}{\partial r}$ and $\frac{\partial V}{\partial h}$ if $V = \frac{4}{3}\pi r^2h$
35. u_E if $u = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} B^2$
36. $\frac{\partial}{\partial x} \left(\frac{1}{\sqrt{2\pi\sigma}} e^{-(x-\mu)^2/(2\sigma^2)} \right)$
37. $\frac{\partial Q}{\partial K}$ if $Q = c(a_1K^{b_1} + a_2L^{b_2})^\gamma$
38. z_x and z_y for $z = x^7 + 2^y + x^y$
39. $\frac{\partial z}{\partial y} \Big|_{(1,0.5)}$ if $z = e^{x+2y} \sin y$
40. $\frac{\partial f}{\partial x} \Big|_{(\pi/3,1)}$ if $f(x, y) = x \ln(y \cos x)$

Problems

41. (a) Let $f(x, y) = x^2 + y^2$. Estimate $f_x(2, 1)$ and $f_y(2, 1)$ using the contour diagram for f in Figure 14.21.
- (b) Estimate $f_x(2, 1)$ and $f_y(2, 1)$ from a table of values for f with $x = 1.9, 2, 2.1$ and $y = 0.9, 1, 1.1$.
- (c) Compare your estimates in parts (a) and (b) with the exact values of $f_x(2, 1)$ and $f_y(2, 1)$ found algebraically.

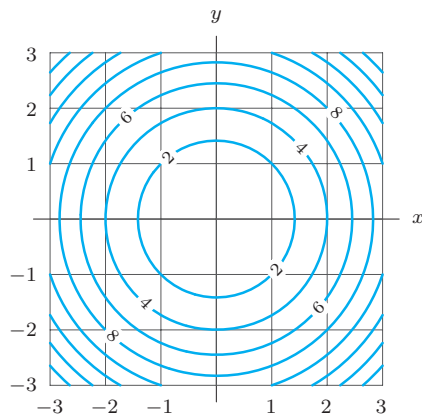


Figure 14.21

42. (a) Let $f(w, z) = e^{w \ln z}$. Use difference quotients with $h = 0.01$ to approximate $f_w(2, 2)$ and $f_z(2, 2)$.

- (b) Now evaluate $f_w(2, 2)$ and $f_z(2, 2)$ exactly.

43. (a) The surface S is given, for some constant a , by

$$z = 3x^2 + 4y^2 - axy$$

Find the values of a which ensure that S is sloping upward when we move in the positive x -direction from the point $(1, 2)$.

- (b) With the values of a from part (a), if you move in the positive y -direction from the point $(1, 2)$, does the surface slope up or down? Explain.

44. Money in a bank account earns interest at a continuous rate, r . The amount of money, $\$B$, in the account depends on the amount deposited, $\$P$, and the time, t , it has been in the bank according to the formula

$$B = Pe^{rt}.$$

Find $\partial B / \partial t$ and $\partial B / \partial P$ and interpret each in financial terms.

45. The acceleration g due to gravity, at a distance r from the center of a planet of mass m , is given by

$$g = \frac{Gm}{r^2},$$

where G is the universal gravitational constant.

- (a) Find $\partial g / \partial m$ and $\partial g / \partial r$.
- (b) Interpret each of the partial derivatives you found in part (a) as the slope of a graph in the plane and sketch the graph.

46. The Dubois formula relates a person's surface area, s , in m^2 , to weight, w , in kg, and height, h , in cm, by

$$s = f(w, h) = 0.01w^{0.25}h^{0.75}.$$

Find $f(65, 160)$, $f_w(65, 160)$, and $f_h(65, 160)$. Interpret your answers in terms of surface area, height, and weight.

47. The energy, E , of a body of mass m moving with speed v is given by the formula

$$E = mc^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right).$$

The speed, v , is nonnegative and less than the speed of light, c , which is a constant.

- (a) Find $\partial E / \partial m$. What would you expect the sign of $\partial E / \partial m$ to be? Explain.
 (b) Find $\partial E / \partial v$. Explain what you would expect the sign of $\partial E / \partial v$ to be and why.
48. Let $h(x, t) = 5 + \cos(0.5x - t)$ describe a wave. The value of $h(x, t)$ gives the depth of the water in cm at a distance x meters from a fixed point and at time t seconds. Evaluate $h_x(2, 5)$ and $h_t(2, 5)$ and interpret each in terms of the wave.

49. A one-meter long bar is heated unevenly, with temperature in $^\circ\text{C}$ at a distance x meters from one end at time t given by

$$H(x, t) = 100e^{-0.1t} \sin(\pi x) \quad 0 \leq x \leq 1.$$

- (a) Sketch a graph of H against x for $t = 0$ and $t = 1$.
 (b) Calculate $H_x(0.2, t)$ and $H_x(0.8, t)$. What is the practical interpretation (in terms of temperature) of these two partial derivatives? Explain why each one has the sign it does.
 (c) Calculate $H_t(x, t)$. What is its sign? What is its interpretation in terms of temperature?
50. Show that the Cobb-Douglas function

$$Q = bK^\alpha L^{1-\alpha} \quad \text{where} \quad 0 < \alpha < 1$$

satisfies the equation

$$K \frac{\partial Q}{\partial K} + L \frac{\partial Q}{\partial L} = Q.$$

51. Is there a function f which has the following partial derivatives? If so what is it? Are there any others?

$$\begin{aligned} f_x(x, y) &= 4x^3y^2 - 3y^4, \\ f_y(x, y) &= 2x^4y - 12xy^3. \end{aligned}$$

Strengthen Your Understanding

In Problems 52–53, explain what is wrong with the statement.

52. The partial derivative of $f(x, y) = x^2y^3$ is $2xy^3 + 3y^2x^2$.
 53. For $f(x, y)$, if $\frac{f(0.01, 0) - f(0, 0)}{0.01} > 0$, then $f_x(0, 0) > 0$.

In Problems 54–56, give an example of:

54. A nonlinear function $f(x, y)$ such that $f_x(0, 0) = 2$ and $f_y(0, 0) = 3$.
 55. Functions $f(x, y)$ and $g(x, y)$ such that $f_x = g_x$ but $f_y \neq g_y$.
 56. A non-constant function $f(x, y)$ such that $f_x = 0$ everywhere.

Are the statements in Problems 57–64 true or false? Give reasons for your answer.

57. There is a function $f(x, y)$ with $f_x(x, y) = y$ and $f_y(x, y) = x$.
 58. The function $z(u, v) = u \cos v$ satisfies the equation

$$\cos v \frac{\partial z}{\partial u} - \frac{\sin v}{u} \frac{\partial z}{\partial v} = 1.$$

59. If $f(x, y)$ is a function of two variables and $g(x)$ is a function of a single variable, then

$$\frac{\partial}{\partial y} (g(x)f(x, y)) = g(x)f_y(x, y).$$

60. The function $k(r, s) = rse^s$ is increasing in the s -direction at the point $(r, s) = (-1, 2)$.
 61. There is a function $f(x, y)$ with $f_x(x, y) = y^2$ and $f_y(x, y) = x^2$.
 62. If $f(x, y)$ has $f_y(x, y) = 0$ then f must be a constant.
 63. If $f(x, y) = ye^{g(x)}$ then $f_x = f$.
 64. If f is a symmetric two-variable function, that is $f(x, y) = f(y, x)$, then $f_x(x, y) = f_y(x, y)$.
 65. Which of the following functions satisfy the following equation (called Euler's Equation):

$$xf_x + yf_y = f?$$

- (a) x^2y^3 (b) $x+y+1$ (c) $x^2 + y^2$ (d) $x^{0.4}y^{0.6}$